

11 (**short**) Figure 8 shows a rigid flywheel with radius R and mass M mounted at one end of a vertical, elastic, and massless shaft with torsional stiffness k . The other end of the shaft is fixed. The angle of rotation of the flywheel from its equilibrium position is α .

- (a) Derive an expression for the natural frequency of the system for torsional oscillations. Determine the natural frequency for $M = 40$ kg, $R = 0.3$ m, and $k = 1.2 \times 10^6$ Nm rad⁻¹. [3]
- (b) If $\alpha = 0$ and $\dot{\alpha} = 200$ rad s⁻¹ at $t = 0$, find the angular amplitude of the subsequent motion. [3]
- (c) In a real system, it is observed that after 3 cycles, the amplitude of the motion has decreased by 95%. Compare the damped and undamped natural frequencies. [4]

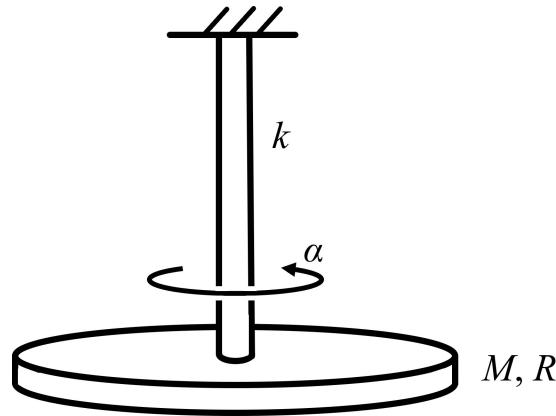


Fig. 8